

SIDDHI MHATRE

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EDUCATION

New York University | Master of Science in Computer Science (GPA: 3.66 / 4.00) Sep 2025 – May 2027
Relevant Coursework: XR: Skill Learning, Computer Graphics, Information Visualization, Artificial Intelligence, VR & AR, HCI

University of Mumbai | Bachelor of Engineering in Computer Engineering (GPA: 3.8 / 4.00) Jul 2019 – May 2023
Relevant Coursework: Data Structures, Machine Learning, Computer Graphics, Artificial Intelligence

TECHNICAL SKILLS

XR & Immersive Interaction: Unity, C#, OpenXR, XR Interaction Toolkit, Meta Quest, AR Foundation, ARCore, WebXR, Photon, Mediapipe
Spatial Visualization & 3D Content: Three.js, React Three Fiber, Mapbox GL JS, Deck.gl, GeoJSON, Blender, Mixamo
Full-Stack, AI & Platform Systems: TypeScript, JavaScript, Python, C++, React, Next.js, Node.js, Express, REST APIs, MongoDB, Redis, Docker, LLM integration, RAG, OpenCV, OCR, Git, Unity Version Control

EXPERIENCE

New York University, Department of Faculty Affairs Jan 2026 – Present
Lead Developer, AI Workflow Dashboarding Platform Brooklyn, New York

- Designed and built a React-based faculty support portal with workflow dashboards for tracking requests, policy resources, approval status, and administrative actions, translating complex policy workflows into a usable interface that reduced manual queries by 88%.
- Built a policy-grounded LLM chatbot using structured retrieval pipelines over institutional documents, improving response accuracy by 92% and reducing query resolution time from hours to 4–5 minutes while supporting human-in-the-loop review.

Jio Platforms Limited (JPL) Sep 2023 – Jun 2025
Software Development Engineer Mumbai, India

- Developed internal HR and enterprise workflow products across onboarding, offboarding, approvals, surveys, and employee lifecycle processes, translating manual operational workflows into structured digital systems.
- Built a Node.js and MongoDB workflow and survey engine with scheduled task flows, structured data capture, real-time reporting, and reusable backend services, replacing legacy SharePoint processes and reducing turnaround time by 85%.
- Implemented supporting platform services including OCR/LLM-based document parsing, queue-based SMS/email notifications, SAML 2.0 authentication with Passport.js, logging, monitoring, and alerting to improve reliability and traceability across production workflows.

Jio Platforms Limited (JPL) Mar 2023 – Sep 2023
Backend Developer Intern Mumbai, India

- Optimized high-volume file upload workflows by improving request handling and backend processing, reducing average latency by 70–80% and supporting faster batch uploads.
- Integrated SAP HANA APIs into a shared microservices architecture and refactored authentication logic, improving reuse across 5 core services and strengthening backend reliability.

PROJECTS

Pit Crew VR: Collaborative Multiplayer F1 Pit Stop | Unity, C#, XR Interaction Toolkit, Photon PUN 2, Mixamo, Meta Quest

- Built a three-player collaborative VR pit stop simulation where users coordinate as Spotter, Jack Operator, and Tire Swapper to complete a timed, five-step tire-change task in a shared virtual environment.
- Implemented networked role selection, scene transitions, synchronized object interactions, avatar animations, action audio, particle effects, and shared game-state logic using Photon PUN 2 RPCs.
- Developed a cinematic replay system with first-person VR views, fixed overview cameras, scripted director-camera movement, fade transitions, and Unity Recorder integration for highlight capture.

Hands of Heritage: Immersive Cultural XR Experience | Unity, C#, Groq, Whisper, Google Cloud TTS, Blender, Mixamo

- Built an immersive XR learning experience that teaches Warli, a traditional Indian folk art through first-person village exploration, NPC-guided cultural storytelling, ritual-based interaction, and hands-on painting on virtual mud walls.
- Designed scaffolded-to-freehand learning flows including shape tracing, motif assembly, ghost-outline guidance, bamboo-brush painting, and personalized drawing prompts generated through conversational AI dialogue.
- Conducted a mixed-methods user study with 12 participants, analyzing task duration, brush-stroke activity, canvas coverage, survey responses, and qualitative feedback to evaluate learning order, usability, comfort, and creative transfer.

Invisible Safety: XR Urban Risk Visualization | Three.js, Deck.gl, Mapbox GL JS, WebXR, ARCore

- Built an AR-based spatial visualization system for exploring hidden urban risks, combining a Deck.gl + Mapbox dashboard with a WebXR AR view that situates infrastructure and safety data in physical space.
- Designed layered geospatial visual encodings for subsurface infrastructure, street-level hazards, risk intensity, data confidence, and incident frequency using color, opacity, thickness, animation, and uncertainty regions.
- Integrated 6 NYC OpenData sources with 2M+ geocoded records spanning sewer networks, 311 incidents, DOT permits, NYPD complaints, Vision Zero collisions, and MapPLUTO footprints.